

Industrial Automation & Control

Week 12 Study Notes + Detailed Assignment Solutions

Energy Savings with Variable Speed Drives + Introduction to CNC Machines

Fast exam revision notes from scratch, plus step-by-step solutions for the uploaded 12-question paper.

What this pack gives you

- concept notes from zero
- high-yield formulas and trends
- short descriptive answers
- one-page revision checklist

What is solved here

- step-by-step reasoning for all 12 uploaded questions
- why each answer is right
- what concept each question tests
- quick answer list for final revision

1. Ultra-short cheat sheet

- For fans and pumps, speed control saves much more energy than throttling or damper control because power demand falls sharply with speed.
- Fan affinity laws: $Q \propto N$, $H \propto N^2$, $P \propto N^3$ where Q = flow, H = head/pressure, P = power, N = speed.
- Pump affinity laws follow the same trend approximately for centrifugal pumps: flow with speed, head with speed squared, power with speed cubed.
- Damper or throttling control wastes energy because it reduces flow by adding resistance instead of reducing machine speed.
- A Variable Frequency Drive (VFD) changes motor supply frequency and voltage, so motor speed changes smoothly and efficiently.
- CNC means Computer Numerical Control: motion and machining actions are controlled by a programmed sequence of coded instructions.
- Point-to-point CNC moves the tool from one location to another; contouring CNC controls the path continuously during cutting.
- Absolute programming measures coordinates from a fixed origin; incremental programming measures each move from the current position.
- Interpolation means generating intermediate positions so the tool follows a straight line or circular path.
- A basic CNC system includes MMI, controller/MCU, PLC logic, servo drives, motors, encoders, spindle drive, and machine mechanics.

2. Energy savings with variable speed drives (VSD / VFD)

- **What a VFD does:** it converts fixed-frequency AC to variable-frequency AC so the induction motor speed can be adjusted.
- **Where it is useful:** fans, blowers, pumps, compressors, HVAC systems, and variable-torque loads.

- **Why energy is saved:** many fluid systems do not need full flow all the time. Instead of running at full speed and wasting energy in a damper/valve, the motor speed is reduced.
- **Duty cycle idea:** if a machine runs for only part of the time or rarely needs full flow, variable-speed operation gives large energy savings over on/off cycling or throttling.

CFM Cubic Feet per Minute, a common unit of airflow rate.

RPM Revolutions per minute, used for rotational speed.

HP Horsepower, a unit of power rating.

2.1 Fan laws and fan control

- **Fan laws:** for geometrically similar operation, flow Q varies directly with speed N , pressure/head H varies as N^2 , and power P varies as N^3 .
- **Big exam point:** power drops very fast when speed is reduced. Example: if speed becomes 80% of rated, power becomes about $0.8^3 = 0.512$ or 51.2% of rated.
- **On/off control:** simple but poor for precise flow control; repeated starting may stress the system and average flow control is rough.
- **Outlet damper control:** the fan still spins near rated speed. Flow is reduced by increasing system resistance, so energy is wasted.
- **Variable fan control:** speed is reduced by VFD so the fan characteristic shifts. The operating point moves to a lower flow and lower pressure with much lower power.

2.2 Pump control

- **Pump characteristic:** relation between head and flow for a given speed. The actual operating point is where the pump curve meets the system/load curve.
- **Throttle control:** a valve is partly closed to reduce flow. This adds losses and wastes power.
- **Variable pump control:** speed is reduced, so the pump curve itself shifts down. Required flow is achieved with lower head and lower power input.
- **Exam comparison:** VFD control is usually superior for variable-flow centrifugal loads; throttling is simpler but less efficient.

2.3 Likely descriptive answer - Why VFD saves energy

- In fan and pump systems, required flow often changes with demand. If conventional damper or throttle control is used, the machine keeps running near full speed and excess energy is dissipated as pressure loss. With a VFD, motor speed is reduced directly. Since flow is proportional to speed and power is approximately proportional to the cube of speed, even a moderate speed reduction gives a large power reduction. Therefore VFD-based control improves efficiency and reduces energy cost.

3. Introduction to CNC machines

- **Numerical control:** machine motions and auxiliary functions are controlled by numbers, codes, and program blocks.

- **CNC:** numerical control with a computer-based controller that stores, edits, and executes the part program.
- **Common machines:** CNC lathe for turning and CNC milling machine or machining center for milling, drilling, boring, tapping, and contouring.
- **Turning:** workpiece rotates while tool feeds along or across the axis.
- **Milling:** cutter rotates while tool/workpiece executes programmed path.

3.1 Motion characteristics

- **Point-to-point control:** only final position matters. Used in drilling, punching, spot welding.
- **Contouring / continuous-path control:** path matters during motion. Used in milling complex profiles, turning contours, and interpolation-based machining.
- **Rapid traverse / rapid feed:** high non-cutting speed used to move between positions quickly.
- **Feed rate:** cutting speed of tool advance, usually programmed separately from spindle speed.

3.2 Coordinate systems and interpolation

- **Absolute system:** every coordinate is given with respect to a fixed origin or datum.
- **Incremental system:** each coordinate is measured from the current point.
- **Linear interpolation:** tool moves in a straight line between two points.
- **Circular interpolation:** tool moves along a programmed arc; radius or center information must be specified.
- **Interpolation parameters:** coordinates, feed, direction, start/end point, and radius/center data depending on the command.

3.3 Part program basics

- **Part program:** ordered sequence of blocks that tell the CNC what to do.
- **Sequential blocks:** each block normally contains one step or instruction set and is executed in order.
- **Typical data in a block:** block number, preparatory function, coordinates, feed rate, spindle speed, tool number, tool offset, and miscellaneous functions.
- **Tool number:** selects which tool station or tool is active.
- **Tool offset / compensation:** corrects for actual tool geometry, wear, and radius so the programmed path gives correct part size.
- **Spindle speed:** rate of spindle rotation. In turning and milling this strongly affects cutting conditions.
- **Thread cutter / threading:** special canned cycles or synchronised spindle-feed motion may be used.
- **Block skip:** allows selected lines to be bypassed during trial or special operation.

3.4 Technical features and architecture of CNC machine

- **Main sections:** MMI/HMI, MCU or CNC controller, PLC section, servo drives, spindle drive, motors, feedback devices, and machine mechanics.
- **MMI:** man-machine interface used for program entry, edit, display, alarm handling, and operation.
- **PLC inside CNC:** handles machine sequencing, interlocks, tool change, coolant, clamps, and auxiliary machine logic.

- **Servo drive:** power electronic unit that receives command signals and drives the axis or spindle motor.
- **Servo motor:** produces controlled motion; AC servo motors are common in modern CNC machines.
- **Encoder:** measures position and/or speed and provides feedback for closed-loop control.
- **Mechanics:** ball screw, guideways, bearings, spindle, gearbox, couplings, and structure.
- **Interfaces:** connect CNC with sensors, actuators, PLC, operator panel, and external devices.

3.5 Drive configurations

- **Open-loop drive:** no position feedback. Simpler and cheaper, often associated with stepper systems in low-cost machines.
- **Closed-loop drive:** command is compared with feedback from encoder/resolver; error is corrected continuously. Better accuracy and dynamics.
- **Why closed loop matters:** machining accuracy, contour following, disturbance rejection, and load variation handling are much better.
- **Axis drive requirement:** enough torque, stiffness, speed range, accuracy, acceleration, and low backlash. Ball-screw load and inertia matter.
- **Spindle drive requirement:** wide speed range, good torque over speed range, speed regulation, reversibility where needed, and good dynamic response.
- **Motor options:** AC servo, DC servo, hydraulic drives in older/special machines. Modern systems mostly use AC servos or vector-controlled spindle motors.

4. Likely exam answers you can write directly

Explain why variable-speed control saves more energy than damper or throttle control. In damper or throttle control, the machine usually continues to run near full speed, and excess energy is lost as pressure drop across the damper or valve. In variable-speed control, the motor speed itself is reduced by a VFD. For fans and centrifugal pumps, flow varies with speed, head varies with the square of speed, and power varies approximately with the cube of speed. Therefore a reduction in speed causes a large reduction in power consumption, so variable-speed control is much more energy efficient.

Differentiate point-to-point and contouring control. In point-to-point control, only the final destination matters; the path taken between the start and end points is not important. This is used in operations such as drilling and punching. In contouring control, the entire path of motion is controlled continuously, and the tool must follow a specified trajectory. This is needed in milling, turning contours, and profile machining.

Differentiate absolute and incremental programming. In absolute programming, all coordinates are measured from a fixed origin. In incremental programming, each coordinate is measured from the present tool position. Absolute programming is easier to verify globally, while incremental programming is convenient for repeated relative moves.

What is interpolation in CNC? Interpolation is the method by which the CNC controller generates intermediate coordinate points between programmed positions so that the tool follows a required path such as a straight line or a circular arc. Linear and circular interpolation are the most common types.

Explain the role of PLC in a CNC machine. The PLC in a CNC machine handles machine-sequence logic and auxiliary operations such as tool change, coolant, spindle start/stop interlocks, clamps, limit switches, and safety logic. The CNC controller handles path control and interpolation, while the PLC handles the machine automation around it.

5. Detailed solutions to the uploaded 12-question paper

Important note: the uploaded paper does not match Lectures 44-47. The uploaded 12-question PDF is about PLC hardware environment, serial communication, and flow-control valves. So this section solves the uploaded paper step by step, while the earlier sections give you the VFD + CNC notes you asked for.

Q1. Optocouplers in PLC I/O units

Correct answer: c -> (i) False, (ii) True

Concept tested: PLC hardware isolation

- Read statement (i): it says optocouplers act like a fuse. That is wrong. A fuse breaks the circuit under excess current; an optocoupler is not a fuse.
- Read statement (ii): optocouplers isolate the CPU from field-side high voltages and currents. That is exactly their purpose.
- So (i) is False and (ii) is True. Therefore option c is correct.

Q2. Serial communication interface

Correct answer: b -> (i) True, (ii) False

Concept tested: Serial vs parallel communication

- Serial communication sends one bit at a time on the line. So statement (i) is true.
- It is not generally faster than parallel communication in the basic sense taught here. Parallel can move multiple bits at once over short distances.
- So (ii) is false. Correct option: b.

Q3. Extra bits X and Z in the RS-232 frame

Correct answer: c -> (i) False, (ii) True

Concept tested: Start and stop bits

- In an asynchronous serial frame, extra bits around the data field mainly indicate start and stop of a character.
- Those bits are not used to directly check whether the message is corrupted.
- Hence (i) is False and (ii) is True. Option c.

Q4. Meaning of bit Y in the shown frame

Correct answer: c -> (i) False, (ii) True

Concept tested: Parity bit reasoning

- The given data is 1100001. Count the number of ones: there are three ones, which is odd.
- If the parity bit Y is added to make the total count even, Y must be 1.

- Therefore Y corresponds to even parity, not odd parity.
- So (i) is False and (ii) is True. Option c.

Q5. Parallel communication interface

Correct answer: a -> (i) True, (ii) True

Concept tested: Parallel communication facts

- Parallel communication is suitable for high speed over short distances because multiple bits are transmitted simultaneously.
- IEEE-488 is indeed a well-known standard for parallel instrument communication.
- So both statements are true. Option a.

Q6. Communication over 100 to 300 m with high rate

Correct answer: c -> (i) False, (ii) True

Concept tested: Choosing the right interface

- RS-232 is not meant for long-distance, high-rate industrial communication over 100 to 300 m with strong noise immunity.
- A 20 mA current loop is much more suitable because current-based signalling works better over longer distance and noisy conditions.
- So statement (i) is false and statement (ii) is true. Option c.

Q7. Why flow control is most important in process industries

Correct answer: b

Concept tested: Importance of flow as manipulated variable

- Flow directly determines how much material or energy enters the process.
- Because of that, changing flow directly influences pressure, level, temperature, and composition.
- That matches option b.

Q8. Valve that controls flow by rotating a ball with a hole

Correct answer: c -> Ball valve

Concept tested: Valve construction and working principle

- A ball valve contains a spherical ball with a bore through it.
- Quarter-turn rotation aligns or blocks that bore with the flow path.
- So the correct choice is ball valve, option c.

Q9. Why installed valve characteristic differs from inherent characteristic

Correct answer: d

Concept tested: Valve characteristic in real plant

- Inherent characteristic assumes constant pressure drop across the valve.
- In the real installation, the pressure drop across the valve changes because the rest of the piping system also shares pressure losses.
- That is exactly option d.

Q10. Fastest actuator in valve operation

Correct answer: a -> Solenoid actuator

Concept tested: Actuator speed comparison

- For simple fast on/off motion, solenoid actuators generally switch quickest.
- Pneumatic and hydraulic actuators give more force and stroke capability, but not usually the fastest small switching action.
- Therefore option a is correct.

Q11. Why high-frequency command signals are not followed accurately by valve stem

Correct answer: b

Concept tested: Dynamic limit of actuator

- The stem and actuator cannot move infinitely fast. They have bandwidth and rate limits.
- So if the command changes too fast, the mechanical system cannot track it accurately.
- This matches option b.

Q12. Final question of uploaded paper

Correct answer: c

Concept tested: Answer visible in the uploaded file

- The visible accepted answer on the last page is option c.
- The extracted text is truncated, so the safest grounded statement is the accepted answer recorded in the paper.
- Hence the answer is c.

6. Final answer list for the uploaded 12-question paper

1) c	2) b	3) c	4) c	5) a	6) c
7) b	8) c	9) d	10) a	11) b	12) c

7. One-page revision checklist

- VFD changes motor frequency and speed.
- Fan and pump affinity laws: $Q \propto N$, $H \propto N^2$, $P \propto N^3$.
- Speed control saves more energy than damper/throttle control.
- CNC = computer-controlled programmed machine motion.
- Lathe -> turning. Milling machine -> milling.
- Point-to-point: destination matters. Contouring: path matters.
- Absolute coordinates use fixed origin. Incremental coordinates use current position.
- Interpolation may be linear or circular.
- Part program blocks may contain tool number, feed, spindle speed, offsets, and coordinates.
- PLC in CNC handles sequencing and auxiliaries; CNC controller handles motion/path control.
- Closed-loop drive uses encoder feedback and gives better accuracy.
- For the uploaded paper: serial sends one bit at a time, parity is error detection only, installed valve characteristic differs because valve pressure drop is not constant.